ME488 Homework 1

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Quarto

Quarto enables you to weave together content and executable code into a finished document. To learn more about Quarto see https://quarto.org.

Running Code

When you click the **Render** button a document will be generated that includes both content and the output of embedded code. You can embed code like this:

```
1 + 1
```

```
[1] 2
```

You can add options to executable code like this

```
[1] 4
```

The `echo: false` option disables the printing of code (only output is displayed).

Chapter 2 Problems

2.1	Basic descriptive statistics / standard error / variance
2.7	Normality + one-sample inference
2.10	Interpreting hypothesis-test output
2.15	Two-sample t-test output and interpretation
2.19	One-sample t-test in an engineering context
2.20	Normality assumption and robustness
2.21	Full one-sample t-test + interpretation
2.23	Two-sample comparison with known, unequal σ's
2.25	Two-sample inference / practical interpretation
2.29	Two-sample engineering experiment

Problem 2.1

Problem Statement:

Computer output for a random sample of data is shown below. Some of the quantities are missing. Compute the values of the missing quantities.

Variable	N	Mean	SE Mean	Std. Dev.	Variance	Minimum	Maximum
Y	9	19.96	NA	3.12	NA	15.94	27.16

Problem 2.7

Problem Statement:

Consider the following sample data:

```
9.37, 13.04, 11.69, 8.21, 11.18, 10.41, 13.15, 11.51, 13.21, 7.75
```

Is it reasonable to assume that this data is a sample from a normal distribution? Is there evidence to support a claim that the mean of the population is 10?

Problem 2.10

Problem Statement:

A two-sample t-test has been conducted and the sample sizes are $n_1 = n_2 = 10$. The computed value of the test statistic is to $t_0 = 2.15$. If the null hypothesis is two-sided, an upper bound on the P-value is:

(a) 0.10 (b) 0.05 (c) 0.025 (d) 0.01 (e) None of the above

Problem 2.15

Problem Statement:

Two-Sample t-Test and Cl: Y1, Y2

```
Two-sample T for Y1 vs Y2
Mean Std. Dev. SE Mean
Y1 20 50.19 1.71 0.38
Y2 20 52.52 2.48 0.55
Difference = mu (X1) - mu (X2)
Estimate for difference: -2.33341
95% CI for difference: (-3.69547, -0.97135)
T-Test of difference = 0 (vs not =): T-Value = -3.47
```

```
P-Value = 0.001 DF = 38
Both use Pooled Std. Dev. = 2.1277
```

(a) Can the null hypothesis be rejected at the 0.05 level? Why?

(b) Is this a one-sided or a two-sided test?

(c) If the hypotheses had been $H_0 : \mu_1 - \mu_2 = 2$ versus $H_1 : \mu_1 - \mu_2 \neq 2$, would you reject the null hypothesis at the 0.05 level?

(d) If the hypotheses had been $H_0 : \mu_1 - \mu_2 = 2$ versus $H_1 : \mu_1 - \mu_2 < 2$, would you reject the null hypothesis at the 0.05 level? Can you answer this question without doing any additional calculations? Why?

(e) Use the output and the t table to find a 95 percent upper confidence bound on the difference in means.

(f) What is the P-value if the hypotheses are $H_0 : \mu_1 - \mu_2 = 2$ versus $H_1 : \mu_1 - \mu_2 \neq 2$?

Problem 2.19

Problem Statement:

The shelf life of a carbonated beverage is of interest. Ten bottles are randomly selected and tested, and the following results are obtained:

(a) We would like to demonstrate that the mean shelf lifeexceeds 120 days. Set up appropriate hypotheses for investigating this claim.

(b) Test these hypotheses using a = 0.01. What are your conclusions?

(c) Find the P-value for the test in part (b).

(d) Construct a 99 percent confidence interval on the meanshelf life.

Problem 2.20

Problem Statement:

A

Problem 2.21

Problem Statement:

A

Problem 2.23

Problem Statement:

A

Problem 2.25

Problem Statement:

A

Problem 2.29

Problem Statement:

A